

Erdős Problem #1083

EP-1083

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $d \geq 3$, and let $f_d(n)$ be the minimal m such that every set of n points in $\mathbb{R}^d$ determines at least m distinct distances. Estimate $f_d(n)$ - in particular, is it true that $f_d(n) = n^{\frac{2}{d}-o(1)}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>